MIL-A-16288D
2 November 1973
SUPERSEDING
MIL-A-16288C
24 November 1965

MILITARY SPECIFICATION

ADAPTERS, COMPRESSED GAS CYLINDER VALVE CONNECTIONS:

GENERAL SPECIFICATION FOR

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers gas cylinder valve adpaters for connecting cylinder valve outlets to regulators, manifolds and other mating parts having different connections.

1.2 Classification. Adapters shall be as specified (see 6.2) in accordance with the applicable specification sheet listed in table I.

2. APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect on date of invitation for bids or request for proposal form a part of this specification to the extent specified herein:

SPECIFICATIONS

Federal

QQ-B-626	- Brass, Leaded and Nonleaded: Rod, Shaped, Forgings, and Flat Products with Finished Edges (Bar and Strip).
PPP-B-601	- Boxes, Wood, Cleated-Plywood.
PPP-B-636	- Boxes, Shipping, Fiberboard.

Military

MIL-S-6872	- Soldering Process, General Specification for.
MIL-B-7883	- Brazing of Steels, Copper, Copper Alloys, Nickel Alloys, Aluminum and Aluminum Alloys.

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STANDARDS

Military

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-129 - Marking for Shipment and Storage.

(Copies of specifications and standards required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids or request for proposal shall apply.

NATIONAL BUREAU OF STANDARDS

Handbook H28 - Screw-Thread Standards for Federal Services.

(Application for copies should be addressed to the superindendent of Documents, Government Printing Office, Washington, DC 20402.)

AMERICAN NATIONAL STANDARDS INSTITUTE, INC.

B57.1 - Compressed Gas Cylinder Valve Outlet and Inlet Connections.

(Application for copies should be addressed to the American National Standards Institute, Inc., 1430 Broadway, New York, NY 10018.)

MACHINERY'S HANDBOOK

(Application for copies should be addressed to the Industrial Press Inc., 200 Madison Ave., New York, NY 10016.)

UNIFORM CLASSIFICATION COMMITTEE, AGENT

Uniform Freight Classification

(Application for copies should be addressed to the Uniform Classification Committee, ATTN: Tariff Publishing Officer, Room 1106, 222 South Riverside Plaza, Chicago, IL 60606.)

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NATIONAL MOTOR FREIGHT TRAFFIC ASSOCIATION, INC., AGENT

National Motor Freight Classification

(Application for copies should be addressed to the American Trucking Associations, Inc., ATTN: Tariff Order Section, 1616 P Street, NW, Washington, DC 20036.)

3. REQUIREMENTS

3.1 Description. The adapters shall consist of (1) an inlet nut, a nipple and a valve outlet connection, or (2) a valve outlet connection and an adapter yoke connection, as applicable, and shall be as specified in the applicable specification sheet and as specified herein.

3.2 Material. Material shall be as specified herein and in the applicable specification sheet. Materials not specified shall be selected by the supplier and shall be subject to all provisions of the specification.

3.2.1 Unless otherwise specified in the applicable specification sheets (see 6.2), the material of the adapters shall be of brass conforming to QQ-B-626, Alloy No. 360, half hard.

3.3 Assembly. Nipple shanks and the portion of the valve outlet connection to which they are coupled shall be either both threaded or both unthreaded. Threaded connections shall be either soldered or brazed; unthreaded connections shall be brazed.

3.3.1 Brazing and soldering. Brazing shall be in accordance with MIL-B-7883; soldering shall be in accordance with MIL-S-6872.

3.3.2 Finishes. All finishes shall have normal machined surfaces except that mating faces shall have a surface roughness not more than 32 microinches, and mating faces for a gas tight seal shall have a surface roughness not more than 16 microinches.

3.4 Performance. The adapters shall show no evidence of leakage, permanent deformation, or rupture when subjected to the test specified in 4.4.2.1.

3.5 Connections. United States nuts, nipples and outlets shall conform to ANS B57.1 and Handbook H28, and shall be as specified in the applicable specification sheet. Dimensions have been omitted from the specification sheets where standard U.S. connection numbers for outlets, nuts, and nipples are applicable.

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3.5.1 Foreign connections. Foreign thread forms shall conform to the Machinery's Handbook.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Acceptability criteria. Adapters which conform to all requirements in the applicable specification sheet and in sections 3 and 5 of this specification, and pass all applicable examinations and tests in section 4 of this specification will be considered acceptable by the Government.

4.1.2 Component and material inspection. The supplier is responsible for insuring that components and materials used are manufactured, examined, and tested in accordance with referenced specifications and standards.

4.2 Classification of inspection. Inspection shall be classified as follows:

(a) Quality conformance inspection (see 4.3).
(b) Inspection of preparation for delivery (see 4.5).

4.3 Quality conformance inspection.

4.3.1 Sampling. Sampling for examination and tests shall be in accordance with MIL-STD-105, inspection level I.

4.3.2 Examination. Samples selected in accordance with 4.3.1 shall be examined as specified in 4.4.1. AQL shall be 1.0 percent defective.

4.3.3 Tests. Samples selected in accordance with 4.3.1 shall be tested as specified in 4.4.2. AQL shall be 1.0 percent defective.

4.4 Inspection procedure.

4.4.1 Examination. The adapters shall be examined for the following defects:

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101. Components or parts not as specified.
102. Threads not as specified.
103. Material not as specified.
104. Dimensions not as specified.
105. Finish not as specified.
106. Assembly not as specified.
107. Marking not as specified.

4.4.2 Tests.

4.4.2.1 Proof pressure. Subject each adapter internally, while the adapter is submerged in a liquid, to an air or nitrogen pressure equal to 1.67 times the service pressure specified on the specification sheet. Non-conformance to 3.4 shall constitute failure of this test.

4.5 Inspection of preparation for delivery.

4.5.1 Quality conformance inspection of pack.

4.5.1.1 Inspection stages. Inspection shall be in two stages as follows:

(a) The first stage shall include inspection of packaging and marking prior to closing the shipping containers.
(b) The second stage shall include inspection of closure and marking of the shipping containers.

4.5.1.2 Units of product. For the purpose of inspection, a unit of product for the first stage of inspection shall be a complete pack with the box top not in place. For the second stage of inspection, a unit of product shall be a completed pack prepared for shipment.

4.5.1.3 Sampling. Sampling for examination shall be in accordance with MIL-STD-105.

4.5.1.4 Examination. Samples selected in accordance with 4.5.1.3 shall be examined for the following defects. AQL shall be 2.5 percent defective.

108. Materials and containers not as specified for level A or B.
109. Boxes not waterproof sealed with tape for level A packaging.
110. Strapping not zinc-coated for level A.
111. Boxes not waterproof sealed with tape for level B packing.
112. Marking illegible, incomplete, or incorrect.

5. PREPARATION FOR DELIVERY

5.1 Packaging. Packaging shall be level A, B, or C as specified (see 6.2).

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5.1.1 Level A. Each adapter shall be packaged in a close-fitting box conforming to PPP-B-636, W6c. After closure the box shall be waterproof sealed by method V in accordance with the appendix to the box specification.

5.1.2 Level B (see 6.3). Each adapter shall be packaged in a close-fitting box conforming to PPP-B-636, type CF, class-domestic, variety SW, grade 125. The boxes shall be closed in accordance with the appendix to the box specification.

5.1.3 Level C. Each adapter shall be packaged in a close-fitting container that will afford protection against deterioration and damage from the supplier to the initial destination.

5.2 Packing. Packing shall be level A, B, or C, as specified (see 6.2).

5.2.1 Level A. The adapters of like description, packaged as specified in 5.1, shall be packed together in quantities not to exceed the size and weight limitation of the boxes, in close-fitting boxes conforming to PPP-B-601, overseas type, style optional. Strapping shall be zinc-coated.

5.2.2 Level B. The adapters of like description, packaged as specified in 5.1, shall be packed together in quantities not to exceed the weight and size limitation of the boxes, in close-fitting boxes conforming to PPP-B-636, V3c. After closure, the boxes shall be waterproof sealed with tape as specified for slotted style boxes in accordance with the appendix to the box specification.

5.2.3 Level C. The adapters of like description, packaged as specified in 5.1, shall be packed to assure carrier acceptance and safe delivery to destination at the lowest rates in compliance with Uniform Freight Classification rules or National Motor Freight Classification rules.

5.3 Marking. Packages and shipping containers shall be marked in accordance with MIL-STD-129. The nomenclature shall include the applicable specification sheet number (see 6.4).

6. NOTES

6.1 Intended use. Adapters specified in the applicable specification sheets are intended for use in adapting valve outlets to regulators and charging manifolds for the same gas, when connections have different threads. Adapters are provided for both U. S. and foreign applications.

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6.2 Ordering data. Procurement documents should specify the following:

(a) Title, number, and date of this specification.
(b) Type of adapter required (see 1.2).
(c) When the material shall be other than as specified (see 3.2.1).
(d) Level of packaging and level of packing required (see 5.1 and 5.2). Level B packaging (see 5.1.2 and 6.3) is intended to provide limited but economical protection and should be specified only when it is determined the adapters will be held in covered storage.

6.3 Packaging. Level B packaging (see 5.1.2) has been developed for use by the Army and Air Force for the specific conditions specified in 6.2. For Navy procurements, details of level B packaging, when required, should be developed at the time of procurement when all conditions are known.

6.4 Specification sheets. This specification contains general requirements for gas cylinder valve adapters. Specific requirements by intended use are covered in separate specification sheets as listed in table I.

6.5 Supersession data. Supersession data is contained in the applicable specification sheets; the following MS standards have been superseded herein: MS39001 through MS39012; MS39220 through MS39223; MS35660, MS35661, MS39274, and MS39275. The following MS standards have been cancelled by previous separate action: MS35662, MS35663, MS35666 through MS35669, and MS39276. The following MS standards are deleted from this specification: MS39272 and MS39273.

Custodians:

Army - ME
Navy - YD
Air Force - 68

Review activities:

Army - MD
Navy - SH, MS

User activities:

Army - AV
Navy - MC, AS

Preparing activity:

Army - ME

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TABLE I

ADAPTERS, COMPRESSED GAS CYLINDER VALVE CONNECTIONS:

Specification Sheet	Title
MIL-A-16288/1	- Adapter, Compressed Gas Cylinder Valve Connections; U. S. Regulator to British Oxygen Valve.
MIL-A-16288/2	- Adapter, Compressed Gas Cylinder Valve Connections; U. S. Regulator to French Oxygen Valve.
MIL-A-16288/3	- Adapter, Compressed Gas Cylinder Valve Connections; U. S. Regulator to Japanese Oxygen Valve.
MIL-A-16288/4	- Adapter, Compressed Gas Cylinder Valve Connections; U. S. Regulator to Dutch Oxygen Valve.
MIL-A-16288/5	- Adapter, Compressed Gas Cylinder Valve Connections; U. S. Regulator to German Oxygen Valve.
MIL-A-16288/6	- Adapter, Compressed Gas Cylinder Valve Connections; U. S. Regulator to British Acetylene Valve.
MIL-A-16288/7	- Adapter, Compressed Gas Cylinder Valve Connections; U. S. Regulator to Dutch Acetylene Valve.
MIL-A-16288/8	- Adapter, Compressed Gas Cylinder Valve Connections; U. S. Regulator to German Acetylene Valve.
MIL-A-16288/9	- Adapter, Compressed Gas Cylinder Valve Connections; U. S. Regulator to French Acetylene Valve.
MIL-A-16288/10	- Adapter, Compressed Gas Cylinder Valve Connections; U. S. Regulator to French Hydrogen Valve.
MIL-A-16288/11	- Adapter, Compressed Gas Cylinder Valve Connections; U. S. Regulator to German & Dutch Hydrogen Valves.
MIL-A-16288/12	- Adapter Yoke, Compressed Gas Cylinder Valve Connections; U. S. Adapters to Dutch, French or German Acetylene Valves.

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MIL-A-16288/13	- Adapter, Compressed Gas Cylinder Valve Connections; Class 200-510 Acetylene.
MIL-A-16288/14	- Adapter, Compressed Gas Cylinder Valve Connections; Class 510-200 Acetylene.
MIL-A-16288/15	- Adapter, Compressed Gas Cylinder Valve Connections; Class 200-520 Acetylene.
MIL-A-16288/16	- Adapter, Compressed Gas Cylinder Valve Connections; Class 500-510 Acetylene.
MIL-A-16288/17	- Adapter, Compressed Gas Cylinder Valve Connections; Class 510-300 Acetylene.
MIL-A-16288/18	- Adapter, Compressed Gas Cylinder Valve Connections; Class 300-520 Acetylene Valve.
MIL-A-16288/19	- Adapter, Compressed Gas Cylinder Valve Connections; Class 520-300 Acetylene.
MIL-A-16288/20	- Adapter, Compressed Gas Cylinder Valve Connections; Class 510-520 Acetylene.
MIL-A-16288/21	- Adapter, Compressed Gas Cylinder Valve Connection; Class 520-510 Acetylene.
MIL-A-16288/22	- Adapter, Compressed Gas Cylinder Valve Connections; Class 580-000 Nitrogen.
MIL-A-16288/23	- Adapter, Compressed Gas Cylinder Valve Connections; Class 000-580 Nitrogen.

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DOCUMENT IDENTIFIER AND TITLE

NAME OF ORGANIZATION AND ADDRESS

CONTRACT NUMBER

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE DOCUMENT CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY DOCUMENT REQUIREMENT CONSIDERED TOO RIGID

3. IS THE DOCUMENT RESTRICTIVE?

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4. REMARKS

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